When Wavelets Meet Dynamic Graph: a Survey in Traffic Forecasting Graph based Models

Nicolò Rossi
Università degli studi di Modena e Reggio Emilia
272179@studenti.unimore.it

Abstract

Accurate traffic forecasting is crucial for efficient transportation management and urban planning. However, it is very challenging to have precise predictions when we talk about complex spatiotemporal dynamics such as traffic data, compounded by unforeseen anomalous events. Graph Neural Networks are widely used for traffic prediction due to their ability to capture correlations between network nodes. This study explores two of these models, STWave [2] and DDGCRN [8] designed to address these challenges through distinct but complementary mechanisms. STWave focuses on mitigating the effects of distribution shifts in traffic time series by employing a disaggregation-fusion framework. On the other hand, DDGCRN addresses the dynamic nature of spatial correlations and the impact of anomalous signals. These two models are compared on a ad hoc created dataset on the Modena Road Network.

1. Introduction

As the connections and technologies between humans increase, it becomes important to know the traffic status of a road so that the traffic flow of cars can be redirected in other directions. With the advent of technology, more and more sensors have been released in the streets to provide useful traffic information. Figure 1 shows an example of the position of some sensors on the Modena road network. Given the observed traffic time series and underlying road networks, traffic forecasting aims to predict a sequence of traffic time series in the future, which benefits daily travel, traffic management, and risk assessment. Despite its importance, traffic forecasting is a challenging task due to the complex nature of traffic data, indeed traffic data exhibit complex temporal patterns influenced by factors such as time of day, day of the week, and special events. But also the presence of dynamic spatial correlations between sensors can increase the difficulty to predict future traffic. And at the end unexpected traffic disruptions,

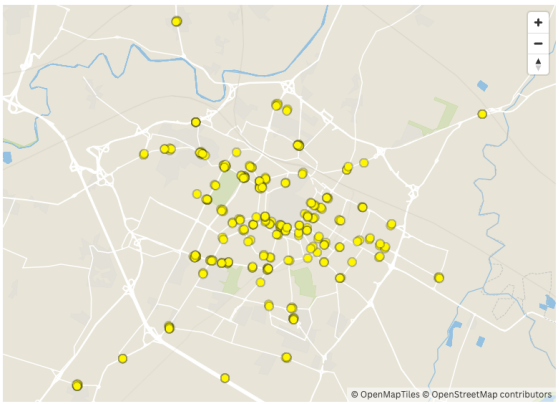


Figure 1. Sensors deployed on the Modena road network

such as accidents or roadwork, can introduce anomalies in traffic data, complicating accurate predictions. Therefore, it has become a pressing need to capture the temporal changes and spatial correlations for accurately forecasting traffic in the future. Early approaches relied on statistical methods like ARIMA [9] and VAR [5]. However, these methods often struggled to capture the complex non-linear relationships in traffic data. With the advent of deep learning, advanced neural networks such as LSTM [1], TCN [7] and Transformer [11] emerged for traffic forecasting. These models can learn complex temporal patterns but often overlook spatial correlations between different points in the road network. In recent years, spatiotemporal graph neural networks (STGNNs) have gained popularity for traffic forecasting. STGNNs leverage the power of graph neural networks (GNNs) to capture spatial correlations between different traffic sensors, modeled as nodes in a graph. By combining GNNs with temporal architectures such as RNNs or CNNs, STGNNs can model both spatial and temporal dependencies in traffic data [4, 6, 10]. Another interesting approach combines knowledge graphs with GCN to provide a knowledge-based model [12].

2. Modena Road Network Graph

2.1. Graph Modeling

An important part of the survey is to produce a correct and true graph of the road network. In the case study of this article we refer to the city of Modena. The graph structure is visible in the Figure 2 and is divided into the following nodes:

- **Road Junction**: Represent a road intersections;
- **Road Section**: Unique identifier of the way in OpenStreetMap;
- **Sensor**: Represent a traffic sensor;
- **Measurement**: Represent a traffic data coming from a sensor in a specific datetime;

and edges:

- *ROUTE*: Indicates that there is a section of road connecting two Road Junction;
- *NEAREST*: Indicates the presents of a traffic sensor in that Road Junction;
- *LOCATED_ON*: Indicates the presents of a traffic sensor in that Road Section;
- *LINK_TO*: Indicates the connection between traffic sensors;
- *HAS*: Indicates a measurement for a traffic sensor;

To create the graph, we followed this pipeline: 1) create the Primal Graph G from OpenStreetMap; 2) Add the sensor nodes and information to G; 3) Preprocessing task. The first two will be discussed in Section 2.2, and the last one in Section 2.3

2.2. Graph Creation

In the Modena graph there are two sources of input data:

1. **Open Street Map** data that's describe the road network, this information are given by a quering the OSM API and save the result in a '.graphml' file which is then imported into the graph.
2. **Sensor information** which provides the localization info of the sensor and flow measurements. Sensor measurements are coming from a file of more than 13 million rows describing the traffic information of Modena from 1 January 2020 to 31 January 2020.

To make the graph more compact and computationally manageable, it was decided to aggregate measurements for time intervals of 15 minutes. Some statistics of the graph are present in Table 1.

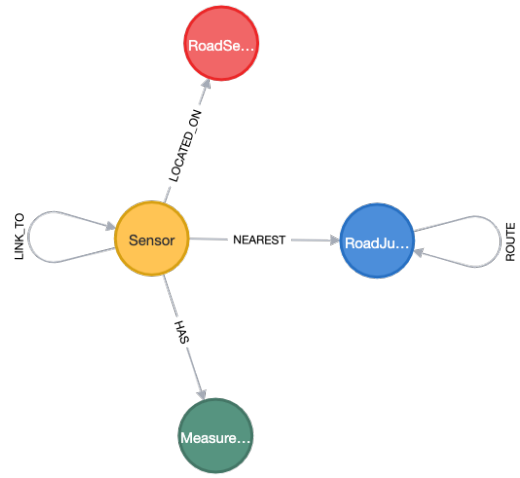


Figure 2. Graph structure of the Modena Road Network

Node	Count
Sensor	259
Road Junction	20963
Road Section	253
Measurement	770784
Tot	792259

Table 1. Statistics of the Modena graph

2.3. Graph Preprocessing

The main task is to make the graph digestible for both models, as will be better described in the Section 3.1, a direct graph $\mathfrak{G} = (V, E, A)$ with the sensors information must be created. To achieve this it is necessary:

1. to create the relationships between sensors;
2. each sensor must have the same number of measurements;

The first step is quite straightforward, given the information on the latitude and longitude of a sensor, we compute the geodesic distance that there is from a sensor i to all other sensors, and create a *LINK_TO* relationship only between the sensors with a distance less than γ meters where we test it with $\gamma = \{300m, 500m, 1000m\}$. The second step is more fuzzy, because in some case, do to error or maintenance, the sensor does not have a single measurement of a particular day, so the number of measurements of every sensors are not equal. For resolve this issue is necessary to find the missing measurements and create them artificially through the use of an interpolation technique. Let us take the example of a sensor a which does not have the measurements of day j, In one day there must be $4 \cdot 24 = 96$ measurements, all measurements of the sensor for all other

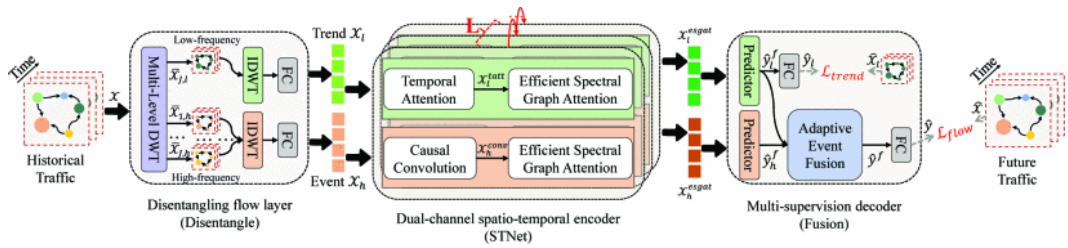


Figure 3. The architecture of STWave. FC: fully-connected layer, DWT: discrete wavelet transform

days in the same time band are taken and an average of the flow values is obtained, These values will then be used to create the measurements in day j.

3. Graph Neural Networks

The following sections describe in more detail the problem of traffic forecasting and provide an overview of the two models examined.

3.1. Preliminaries & Problem Definition

Given the real-world traffic road network and the deployed sensors that record traffic information on the traffic road, we use a directed graph $\mathfrak{G} = (V, E, A)$ to predict traffic, where V is the set of sensors. E is the set of edges between neighboring sensors on the traffic road network, and $A \in \mathbb{R}^{NxN}$ corresponds to the adjacency matrix of $\mathfrak{G}$. The traffic forecasting problem aims to forecast the future traffic on the traffic road network through the known historical traffic data recorded by the deployed sensors. Take, $x_t^i \in \mathbb{R}^C$ where $C = 1$, represents the traffic flow of the *i*th sensor on the traffic network at time step t, and $X_t = [x_t^1, ..., x_t^i, ..., x_t^N]^T \in \mathbb{R}^{NxC}$ represents values of all sensors on the traffic network at time step t.
Given historical T_1 time slices traffic data $\mathcal{X} = \{X_1, ..., X_{T_1}\} \in \mathbb{R}^{T_1xNxC}$ of all sensors and the graph $\mathfrak{G}$ of traffic network, the purpose of traffic forecasting is to learn a function $\mathcal{F}$ to forecast the traffic data of all sensors in the future T_2 time slices, namely $\hat{\mathcal{Y}} = \{\hat{Y}_{(T_1+1)}, ..., \hat{Y}_{(T_1+T_2)}\} \in \mathbb{R}^{T_2xNxC}$ and its ground truth is denoted by $\hat{\mathcal{X}} = \{X_{(T_1+1)}, ..., X_{(T_1+T_2)}\} \in \mathbb{R}^{T_2xNxC}$. The task can be formulated as:

$$\{\hat{Y}_{(T_1+1)}, ..., \hat{Y}_{(T_1+T_2)}\} = \mathcal{F}_\Theta(\{X_{(T_1+1)}, ..., X_{(T_1+T_2)}\}, \mathfrak{G})$$

where Θ denotes the learnable parameters of the model.

3.2. STWave

STWave [2] is an innovative disaggregation-fusion framework designed to address the issue of distribution shift in traffic time series. The architecture of STWave can be seen in Figure 3 .Instead of modeling complex traffic data end-to-end, STWave adopts a four-step approach:

- **Disaggregation**: STWave utilizes multi-level discrete wavelet trasform(DWT) to decomposes traffic time series into two components: **Trends** that represents stable, long-term patterns in the data; **Events** represents short-term fluctuations and abrupt changes.

- **Dual-channel spatio-temporal encoder**: STWave uses temporal attention and causal attention on trends and events to capture the stable and fluctuating temporal changes.

- **Spatial Correlations Extraction**: To model spatial correlations, STWave uses an efficient spectral graph attention network (ESGAT). ESGAT incorporates two different part: a novel query sampling strategy to gain information from neighbors sensors and some active sensors as sparse queries to absorb global information on the graph; and wavelet-based positional encoding to efficiently and effectively capture global spatial dependencies.

- **Multi-supervision decoder**: Given the learned representation of historical trends and events in the future, STWave utilizes two predictors to forecast trends and events in the future, and then uses an adaptive event fusion module to derive the future traffic.

3.3. DDGCRN

DDGCRN (Decomposition Dynamic Graph Convolutional Recurrent Network) [8] is another innovative model designed to tackle real-world traffic forecasting challenges. The architecture can be seen in Figure 4.DDGCRN is composed by:

- **Dynamic graph generation**: Instead of relying on predefined or adaptive adjacency matrices, DDGCRN dynamically generates graph structures based on time-varying traffic signals, making it applicable to any scenario lacking of prior knowledge. To generate dynamic graphs, DDGCRN uses temporal and spatial embeddings to create spatiotemporal embeddings. These embeddings are combined with dynamic signals extracted from traffic data to produce dynamic graph

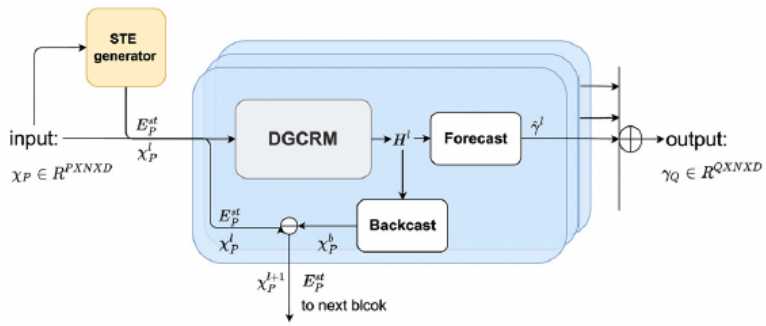


Figure 4. The architecture of DDGCRN

embeddings, which are then used to generate the dynamic graph.This approach enables the model to capture complex spatial dependencies that evolve over time.

- **Node adaptive parameter learning**(NAPL): the DDGCRN is optimized through a module that's enable to learn a unique traffic pattern for each node.
- **Dynamic graph convolution recurrent module**(DGCRM): After creating the spatiotemporal embeddings, they are pass to a DGCRM module composed of DGCRUs(dynamic graph convolution gated recurrent unit) that extract the spatio-temporal characteristics from the traffic signal sequences.
- **Residual decomposition and multi-step prediction**: The DGCRM has two outputs: a forecasting output that is summed to form the final prediction and a backcasting output that is the reconstruction of the input signal. Than through residual decomposition, the information of the input signal that has been learned is removed, and only the part that have not been learned is reserved for modeling in the next block.

3.4. Comparison of STWave and DDGCRN

Both STWave and DDGCRN represent traffic forecasting models that address the limitations of traditional approaches. While they share the goal of improving forecast accuracy, they employ different mechanisms and focus on distinct aspects of the problem:

- **Handling distribution shifts:** STWave focuses on mitigating the issue of distribution shifts in traffic time series by disaggregating data into trends and events. DDGCRN does only handling anomalous signals to improve forecast accuracy.
- **Modeling spatial correlations:** STWave uses ESGAT to efficiently capture global spatial dependencies. DDGCRN dynamically generates graph structures to capture evolving spatial relationships, eliminating the need for prior knowledge of road connections.

4. Experiments

In this section we will show in the first part the results of the performance of the two models in the Modena road network, and then we will show some experiments carried out on the models.

Metrics: In this paper, three widely used metrics are used for all tasks, namely, Mean Absolute Errors (MAE), Mean Absolute Percentage Errors (MAPE), and Root Mean Squared Errors (RMSE).

Hyper-Parameter Settings: STWave is trained with an Adam optimizer for 200 epochs with a batch size of 64 and a learning rate of 0.001, meanwhile DDGCRN is trained with Adam optimizer for 300 epochs with a batch size of 64 and a learning rate of 0.003, also DDGCRN has a early stopping procedure when the model didn't improve for 15 consecutive epochs. Both models take in input 4 past measurements of a sensors and output the next 4 predicted measurements.

Baseline: We also compare the two models with Historical Average (HA) [3] as a baseline model that simply utilizes average value of history to predict the future.

4.1. Perfomance Comparisons

The results are reported in Table 2. We can see that DDGCRN outperforms STWave in every metric. Both models perform much better than HA. The results of the table show that DDGCRN's approach is best suited to the task of forecasting traffic. A clearer display of results can be seen in the Figures 5b and 5c.

Methods	MAE	RMSE	MAPE
HA	14.7804	52.1398	0.5300
STWave	8.1070	22.4635	0.3665
DDGCRN	**5.7676**	**19.8603**	**0.3264**

Table 2. Comparison of STWave, DDGCRN and baselines. **Bold**: best

4.1.1 *Graph Connectivity*

The purpose of this section is to analyze the impact of different graph structures on the performance of the model. The analysis was done only on STWave because DDGCRN does not take into account the graph structure but creates a dynamic graph based on traffic data. To change the graph structure we modify the maximum geodesic distance for creates the *LINK_TO* edge, as explained in section 2.3. The values chosen for the case study was:

1000: It takes a maximum of 1000 metres to establish the relationship between two sensors
500: It takes a maximum of 500 metres to establish the relationship between two sensors
300: It takes a maximum of 300 metres to establish the relationship between two sensors

In Table 3 and in the Figure 5a some graph properties are shown. An interesting properties of these graphs can be seen in Table 3. Although the graph with a γ equal to 1000 at less Weakly Connected Components(WCC) than the graph with γ equal to 500, as it would be fair since the more the γ increases the more you should create connections between nodes, the Average Betweenness Centrality(ABC) is highest in the graph with a γ equal to 500. This shows that in some cases even if the nodes are part of different communities and are less densely connected, on average the number of shortest paths(therefore the importance) that pass through those nodes is greater than in a more densely connected graph. The model results are display in Table 4, we can see that the best results are with γ equals 1000 meters, this shows that a more densely connected graph makes it possible to represent with greater coherence the different spatial interconnections between sensors and thus predict in a more truthful way.

5. Conclusion

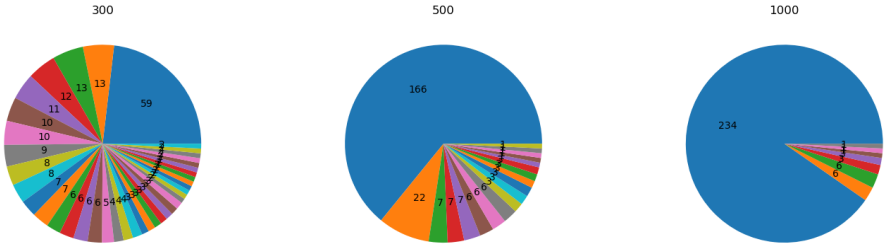
Traffic forecasting remains an active and challenging research area. This paper presents a small view of the world of traffic prediction models. The DDGCRN model with the dynamic graph generation approach and multi-step prediction outperform STWave, this means that a reduces reliance on prior knowledge, further enhances the versatility of spatio-temporal graphs. But there is still room to improve DDGCRN and STWave. First of all DDGCRN simply decomposed traffic signals into normal and abnormal signals meanwhile STWave uses a more interesting concept of trends and events. STWave is unable to achieve the same forecast performance of DDGCRN, this could be due to too much dependence on the graph structure.
Anyway, other models not seen in this study could bring better results. For example, a model that also takes into account the input of events which might be taking place in the city, such as trade fairs or social events, could further improve predictions.

Threshold(m)	ADC	WCC	ABC
1000	64.00	8	433.55
500	22.76	19	610.89
300	12.28	40	46.58

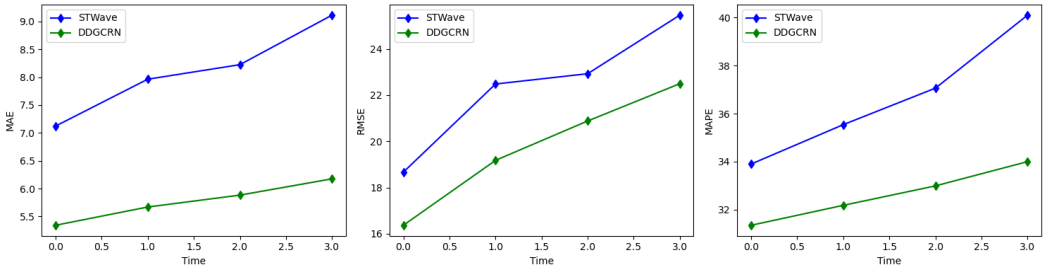
Table 3. Statistics on the different types of graph according to geodesic distance. ADC: Average Degree Centrality, WCC: Weakly Connected Components, ABC: Average Betweenness Centrality

γ(m)	MAE	RMSE	MAPE
300	8.9734	23.6394	0.457
500	9.3433	24.9289	0.408
1000	**8.1070**	**22.4635**	**0.3665**

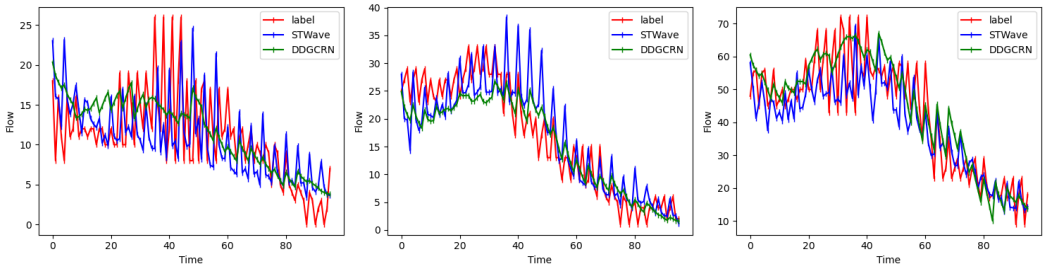
Table 4. Comparison of STWave with different geodesic distance. **Bold**: best



(a) Distribution of Weakly Connected Components with different γ



(b) Prediction for each time slice



(c) Comparison in 1-day measurements

Figure 5. Some visualization results

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